

DISCOVERY PISCINE

FINAL WEEKEND PROJECT

Python Edition

01 | Introduction

Summary: In this final project, you will combine everything you have learned over the week to build a fully playable, visually engaging text-based game.

What this module will show you:

You will apply variables, conditions, loops, lists, and dictionaries to build a comprehensive, interactive program from scratch. This is your chance to be creative and show off what you've learned!

02 | Exercise 00: Tic-Tac-Toe

Directory: `ex00/` | File: `tic_tac_toe.py`

- Create a program called `tic_tac_toe.py` and ensure it is executable.
- When executed, the script must prompt the user to enter a name for Player 1 and a name for Player 2.
- You must design and implement a suitable data structure (like a list or dictionary) to store and update the state of the 9 spaces on the board.
- The main game must run continuously without exiting. It should only stop and exit when a player wins, the board is full (a draw), or if a user inputs `STOP`.
- Prompt the user to input their move. Your program must automatically track whose turn it is, process the input, and verify if the move is valid (e.g., prevent a player from overwriting a space that is already taken).
- Keep a running tally of how many times Player 1 and Player 2 have won across multiple rounds.

Visual Requirements:

In this project you have full creative freedom to make your tic-tac-toe the most beautiful one ever seen, you are free to use any tools or frameworks you wish to.

To provide you a starting point, the example below represents the **minimum requirement** for your interface layout. Your output must be clean and visually distinct. You must draw a proper game board and interface.

```
student@42berlin: ~/tic_tac_toe

$> ./tic_tac_toe.py

  _____
 |                                     |
 |          ASCII ARENA: TIC-TAC-TOE          |
 |                                     |
 |_____|

Player 1, what is your name?: Alice
Player 2, what is your name?: Bob

      CURRENT SCORE
[ Alice: 0 ]      [ Bob: 0 ]

  _____
 | 1 | 2 | 3 |
 |___|___|___|
 | 4 | 5 | 6 |
 |___|___|___|
 | 7 | 8 | 9 |
 |___|___|___|

Alice's turn!
Choose a space (1-9) or type 'STOP': 5

  _____
 | 1 | 2 | 3 |
 |___|___|___|
 | 4 | X | 6 |
 |___|___|___|
 | 7 | 8 | 9 |
 |___|___|___|

Bob's turn!
Choose a space (1-9) or type 'STOP':
```

03 | Submission & Presentation

- You must have the `discovery_piscine` folder at the root of your home directory.
- Inside the `discovery_piscine` folder, you must have a folder named `final_project`.
- Inside the `final_project` folder, you must have an `ex00` folder.
- The `ex00` folder must contain the file requested in the assignment (`tic_tac_toe.py`).

PRESENTATION

This project will **not** be peer-evaluated. You will present your game live to the entire group. You must be prepared to demonstrate your working game and confidently answer questions regarding your data structures, logic, and implementation choices. **Ensure you fully understand the code you have written!**